

Rohan Devraj

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EDUCATION

Georgia Institute of Technology | Atlanta, GA

August 2023 - December 2027

B.S. in Industrial & Systems Engineering, Minor in Computer Science

GPA: 3.89

- *Concentrations:* Operations Research, AI & Machine Learning
- *Coursework:* Probability, Adv. Optimization, Data Structures & Algorithms, Stochastics, Databases, AI

EXPERIENCE

Apple

Jun. 2026 - present

Industrial Engineering Intern

Cupertino, CA

- Developing a Python/SimPy-based simulation engine to enable data-driven prediction and analysis of manufacturing processes, leveraging open-source tools to reduce reliance on proprietary simulation software.
- Designing an analytical framework for supply chain resilience, identifying key indicators and datasets, and developing weighted evaluation methodologies to support data-driven investment decisions.

Optum Technology (UnitedHealth Group)

Jun. 2025 – Aug. 2025

Data Engineering Intern

Minneapolis, MN

- Developed a Spark-based Python library deployed in Azure Databricks to compute AHRQ Quality & Patient Safety measures from ICD-10 healthcare data, replacing legacy SAS systems and saving \$36K annually.
- Reduced processing time by 90% (10 hrs to less than 1 hr) by optimizing with Pandas UDFs and Databricks.
- Built validation pipelines to benchmark Spark outputs against legacy SAS logic and accelerated health data modernization through fast, in-place cloud processing.

Space Systems Optimization Lab at Georgia Tech

Aug. 2024 – May 2026

Undergraduate Researcher

Atlanta, GA

- First-author research on multi-scale path planning for planetary rovers & presenter at AIAA ASCEND 2026.
- Developed a three-layer planetary rover routing framework integrating A* search, Mixed-Integer Linear Programming (Pyomo/Gurobi), and stochastic local replanning to optimize multi-site navigation under uncertain terrain conditions (advised by Dr. Koki Ho).
- Reduced replanning from 52K+ variables and 62K+ constraints to a hierarchical model solved in under 7 seconds.
- Designed an adaptive local planning framework using terrain interpolation and stochastic replanning for obstacle avoidance under partial observability; awarded Georgia Tech's President's Undergraduate Research Travel Award.

Carnegie Mellon Department of Statistics & Data Science

Jun. 2024 – Jul. 2024

Summer Undergraduate Research Fellow | stat.cmu.edu/cmsac/sure/2024/showcase/

Pittsburgh, PA

- Nationally selected (1 of 18) for a competitive research program in data science applied to healthcare analytics.
- Developed regression and machine learning models (linear, lasso, ridge, decision trees) to analyze the impact of adult health behaviors on child mortality and low birthweight at the county level.
- Evaluated and compared models using 10-fold cross-validation; selected linear regression for its 5% lower root mean squared error (RMSE), enhancing low birthweight prediction accuracy.
- Conducted 10+ analyses on maternal healthcare disparities and applied k-means clustering to identify distinct demographic and behavioral risk groups.

LEADERSHIP & PROJECTS

GT Autonomous & Connected Transportation Lab

Undergraduate Research Assistant

- Formulated a multi-objective traffic assignment model in Python using Pyomo and Gurobi to optimize routing across zone-based transportation networks, balancing travel time, congestion, and safety across diverse zones.

Georgia Tech Alumni Association Angel Network

Senior Associate | gtangelnetwork.com

- Lead Georgia Tech's first angel investment network, sourcing 20+ startups from a 100+ deal pipeline and supporting over \$6.5M in connected capital.

Georgia Tech - H. Milton Stewart School of ISyE

Engineering Optimization Tutor

- Tutor 70+ students in optimization topics like linear programming, MILP, solution methods, and network models.

SKILLS

Programming/Tools: Python, SQL, Java, R, Excel, PySpark, Azure Databricks, C, SAS, Git

Libraries & Modeling: Pandas, NumPy, ggplot2, dplyr, AnyLogic, Simio, Gurobi, Pyomo, GurobiPy, CVXPY